SANGAM KUMAR

AI Platform Engineer | Agent Runtimes, LLM Gateways & Production AI Systems

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- Engineer who builds the infrastructure AI products sit on, the **agent runtimes, LLM gateways and cost controls**, and also ships the AI features that run on top of it.
- **Over 7 years** of backend and distributed systems work in Ruby, Go and Python across AWS, GCP and Azure, so the AI systems I build are ones I can also run, debug and keep within budget.
- Open to relocation and remote roles across APAC and EMEA.

TECHNICAL LEADERSHIP

- Reviewed **over 1,400 pull requests across 60+ repositories** so far, including open source, holding steady at around 150 a year since 2020. Have been a regular reviewer for 16 engineers, and am first reviewer for two on the workspace platform today.
- Wrote **6 architecture RFCs and design documents**, covering the shutdown of legacy infrastructure, public API design, container registry migration and candidate IDE tooling.
- Write the **RCAs for P1 and P2 customer escalations**. Part of the on-call rotation, and handle incident response and postmortems.
- Own production releases and phased rollouts, usually **30%, then 50%, then 100%** of traffic, with clear fallback and abort points agreed beforehand.
- On the **system design interview panel** and work with the engineering manager on keeping the hiring bar consistent. Also built the agent-based triage tooling the team now uses on production issues.

EXPERIENCE

HackerRank ***Jul 2022 – Present***

Senior Software Engineer *Bengaluru, India*

- Led the launch of ADE, our agentic development environment, where a candidate gets an editor, a terminal and an agent chat in one workspace, with a coding agent running against the **Anthropic API** through our LLM gateway. Owned the agent server lifecycle, the ingress, **minting a fresh credential per session**, tool provisioning and versioned deployments, across four repositories.
- Brought in **LiteLLM** as our LLM proxy and added **per-tenant budget, TPM and RPM limits** on top of it, so one team or feature could not eat into everyone else's quota. Later added **Portkey** metadata to the requests so spend could be attributed per feature, and tightened up its release pipeline. Also wrote scripts that deliberately try to exhaust a budget, to confirm the limits actually hold.
- Designed and launched the **Prompt Engineering** question type from nothing to GA, covering the backend, question creation, the candidate experience, the checker programs and localisation. Followed it with the **RAG** and **Agent** modules built in Python with OpenAI and LlamaIndex, along with an OpenAI tool runtime and GPU-backed images.
- Keep the AI assessments working as models move on. Upgraded the environments to newer OpenAI model versions and re-validated the RAG questions against them, since a question that scores correctly on one model does not always behave the same way on the next one.
- Shipped the in-workspace AI assistant along with the Copilot and Continue integrations, gated to customers on the AI add-on. Wrote the RCA when the assistant turned itself on for a question type it was not meant for, and fixed the flag handling behind it.
- Built the agent-based triage tooling the team uses now. These are reusable skills that pull workspace state and support context together, so a production escalation can be investigated from logs and telemetry without starting from scratch each time.
- On the platform side, our DevOps product ran on its own provisioning stack that was hard to scale, so I moved it onto the shared multi-cloud Workspace Service and then retired the old one, shutting down an EKS cluster, a managed Redis, 9 cron jobs and 6 workers with no data migration. Also shipped the v3 public Projects API and built `environment-service` in Go, which manages **60+ assessment environments**.
- Fixed duplicate event and transaction race failures in the workspace event pipeline using unique jobs and distributed locks, and traced a weeks-long provisioning slowdown to a single NAT address that was dropping traffic silently. Built the Prometheus and Redash dashboards the team uses for platform SLAs.

HackerRank ***Jan 2021 – Jun 2022***

Software Development Engineer II *Remote*

- Scaled the DevOps assessment platform to run **1,000 servers at a time**. Added a Redis lock so warmup jobs stopped stepping on each other, and used gh-ost for MySQL schema changes so we did not need downtime for them.
- Launched DevOps Assessment 2.0 with a shared interactive terminal for live interviews, after which **usage grew by 500%**. Sent patches upstream to `ttyd` and moved session recording from `ttyrec` to `asciinema` so playback worked properly in reports.
- Led the Flutter stack delivery and added assessment support for AWS, Git, Apache, Terraform and Docker. Handled the Rails 5 to Rails 6 migration for the DevOps modules. Helped take HackerRank's **DLI to 4.3+** and improve NPS.

HackerRank ***Jul 2019 – Dec 2020***

Software Development Engineer *Bengaluru, India*

- Scaled the DevOps assessment tool to handle **900% more concurrent candidate sessions**, owning the EC2 provisioning path, the Packer machine images and logging. Moved its workers and cron jobs onto **Kubernetes**.
- Set up the **SLA capture and monitoring system**, which became the numbers the team tracked for quality and performance, and built a script execution service used for re-scoring and running tests.

HackerRank ***Jan 2019 – Jun 2019***

Software Development Engineer Intern *Bengaluru, India*

- Moved old Backbone pages to React (contests, leaderboards, Hourrank, 101Hack) and built the shared UI icons library, which generated React components from SVG files automatically using SVGR and Gulp.

Google Summer of Code ***Apr 2018 – Aug 2018***

Student Developer *Remote*

- Worked with coala.io on their language-independent static analysis tool. Wrote three new Bears for Ruby and Crystal using Brakeman, Fasterer and Ameba, and improved pytest coverage in the core.

SELECTED WORK & OPEN SOURCE

- Open source: sent patches upstream to `ttyd` for terminal presence and keepalive, wrote the Ruby and Crystal Bears for `coala`, and fixed caching and memoization bugs in `rspec-tracer`.
- Worked on language server performance in candidate IDEs. Added `clangd` so C++ gets proper completion and debugging, and benchmarked `ruff` against `pylint` to fix how slow the Python editor felt.
- Launched Playwright as a QA assessment skill, including the runtime, the images and a flag-based rollout, along with work on the Data Science, Jupyter and full-stack environments.
- On the security side, rotated container registry credentials across 5 repositories, cleared pending Dependabot vulnerabilities in Go and Ruby services, and added the tagging needed for compliance audits.

EDUCATION

National Institute of Technology ***Aug 2015 – May 2019***

Bachelor of Technology in Computer Science and Engineering *Sikkim, India*

SKILLS

AI Engineering	LLM gateways (LiteLLM, Portkey), Anthropic & OpenAI APIs, agentic runtimes and coding agents, RAG, MCP, prompt engineering, guardrails, token and cost budgeting (TPM/RPM), LlamaIndex, evaluation and model drift testing
Languages	Ruby, Go, Python, TypeScript, JavaScript, C++, Java, Bash
Frameworks	Ruby on Rails, ActiveRecord, RSpec, Gin, FastAPI, Node, React, Next.js, Flutter
Data	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch/OpenSearch, Redshift
Infrastructure	AWS, GCP, Azure, Microsoft Entra, Kubernetes, Docker, Terraform, Helm, Packer, Jenkins, GNU/Linux
Observability	Grafana, Prometheus, OpenTelemetry, NewRelic, Kibana, Redash, CloudWatch